SOLAR AGRICULTURAL AGENCY
Answer Key

SSS-C1-CASE02

Lunar Greenhouse

Campaign 1 · Case 02 · Shackleton Crater, Lunar South Pole

v1.0
VALIDATION BUILD

COMPLETED-EXEMPLAR REQUIREMENT

Every keyable field is completed below. Tasks 1 and 2 are non-keyable and remain omitted without renumbering.

TASK 03 3 · Model the pollination sequence

WORD BANK **fruit set · physical agitation · viable pollen in anthers · fertilization · pollen tube growth · pollen reaches stigma**

1	viable pollen in anthers	WORKING
2	physical agitation	FAILED STEP
3	pollen reaches stigma	DOWNSTREAM BLOCKED
4	pollen tube growth	DOWNSTREAM BLOCKED
5	fertilization	DOWNSTREAM BLOCKED
6	fruit set	DOWNSTREAM BLOCKED

COMPLETED EXEMPLAR The first failed event blocks later events even when downstream structures remain capable.

MODEL EXPLANATION

Each arrow means the earlier event makes the later event possible. The sequence distinguishes pollen transfer from fertilization and fruit set.

TASK 04 4 · Identify the failed step

FAILED STEP

Step 2 — **physical agitation** that releases pollen from the anthers.

OBSERVATION 1

Abundant, apparently viable pollen remains undisturbed on anthers while stigmas remain clean.

OBSERVATION 2

Air is nearly still; no insects, fans, or implemented manual/mechanical pollination plan are present.

DOWNSTREAM RESULT

No stigma contact, pollen-tube growth, fertilization, or fruit set; flowers drop.

TASK 05 5 · Test the competing explanations

Possible diagnosis	Completed exemplar
Lunar regolith is toxic to fruit development.	**Weakened.** Roots, leaves, stems, nutrients, and moisture are healthy; broad toxicity does not explain pollen staying on anthers.
The light spectrum lacks ultraviolet light needed for fruiting.	**Weakened.** Flowers form normally and the case reports full-spectrum light; undisturbed pollen is the direct bottleneck.
No effective pollen release and transfer is occurring.	**Supported.** Pollen is viable but undisturbed, stigmas are clean, air is still, insects are absent, and no plan was implemented.
Carbon dioxide is too high for tomatoes.	**Weakened.** The sensor treats the value as compatible with growth, and it does not explain the flower-level pattern.

TASK 06 6 · Diagnose and reject an alternative

DIAGNOSIS

The crop fails at pollen release and transfer. The sealed habitat supplies no effective flower agitation, so pollen remains on anthers instead of reaching the stigma. Later reproductive steps cannot proceed.

REJECTED ALTERNATIVE

Regolith toxicity is tempting in a lunar setting, but healthy roots and vegetative growth weaken it. It also does not explain undisturbed pollen on normal flowers.

TASK 07 7 · Claim-Evidence-Reasoning

CLAIM

The tomatoes do not set fruit because the greenhouse omitted effective pollination, especially physical agitation that releases pollen and enables stigma contact.

EVIDENCE

Plant examination shows mature pollen still on anthers and clean stigmas. Sensors show nearly still air and no fans. Design records leave pollination unresolved, and no insects are present.

REASONING

Healthy flowers can produce viable pollen without completing reproduction. Pollen must be released and reach a receptive stigma before it germinates and grows a pollen tube. Because no effective agitation occurs, the sequence stops before stigma contact, so fertilization and fruit set cannot follow.

TASK 08 8 · Design a reliable pollination support

DESIGN
Use scheduled, targeted flower-cluster vibration, with a manual backup during equipment outages.

CRITERION
Consistently release pollen and produce verified stigma contact/early fruit set across open clusters.

CONSTRAINT
Operate in a sealed habitat with limited power, mass, crew time, and no animal pollinators, without damaging flowers.

MECHANISM CHECK
Restore Step 2, physical agitation. Verify pollen release during operation, then monitor stigma contact and early ovary swelling.

Accepted variation: A verified manual schedule or carefully controlled airflow is defensible when the response restores pollen release/transfer, protects flowers, and includes a measurable check.

TASK 09 9 · Exit ticket

COMPLETED EXEMPLAR
First test pollen release and stigma contact because flowers and viable pollen are already confirmed. Inspect whether pollen remains trapped on anthers and whether receptive stigmas receive grains before and after controlled agitation. If pollen already reaches the stigma, move downstream to germination, pollen-tube growth, or fertilization.

SCORING CAUTION
Do not require exact wording outside the Task 3 word bank. Do not score game points, speed, rapport, or optional exploration.

Dimension	Secure evidence
Evidence	Specific observations from more than one source.
Mechanism	Correct sequence and first failed step.
Reasoning	Evidence linked to process and alternative rejected.
Completion	Tasks 3–9 complete, including criterion and constraint.